

Nathan Sarkozy

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OBJECTIVE

I am a Computer Engineering student seeking an internship in embedded systems or hardware design. Passionate about developing integrated digital and physical systems through hands-on experience with microcontrollers, sensors, and FPGA-based projects.

EDUCATION

B.S Computer Engineering

California State University, Long Beach

Expected Graduation: Spring 2027

Major GPA: 3.90

Presidents Honor List

A.S Computer Science

Modesto Junior College

Graduation: Spring 2023

Cumulative GPA: 3.10

RELEVANT PROJECTS

FPGA – Based Digital Compass with Sensor Integration | Co-Embedded Software Engineer

- Developed real-time digital compass on Nexys A7 FPGA with Pmod CMP32 magnetometer and accelerometer
- Implemented Verilog modules for sensor data acquisition, tilt compensation, and heading computation while utilizing I2C and SPI protocols for sensor communication

ESP32 – Digital Golf GPS Range Finder |

- Developed and assembled ESP32-based handheld device with NEO-6M GPS module for real time yardage distances
- Displayed current hole, back, front, and middle distance to pin, and possible hazards on course

Line Follower FSM | Lead Hardware Engineer

- Developed line following robot using a TM4C123GH6PM Microcontroller with an 8-channel IR reflectance sensor array to detect surfaces. Involved 3D printing servo-controlled gate that use IR sensors to detect incoming object.
- Designed a finite state machine to interpret sensor inputs and control motor outputs for left, right and straight movements

EXPERIENCE

Computing Consultant | Escalon High School

August 2020 – May 2022

- Performed troubleshooting for network connectivity, hardware, and virus problems
- Supported staff and faculty in computer technology and security

Lead Golf Technician | Dick's Sporting Goods

September 2021 – Present

- Perform club fittings, repair, and maintenance for golf equipment including drivers, irons and putters.
- Utilize launch monitor technology to analyze swing data and recommend optimal equipment configurations
- Conduct grip replacements, shaft adjustments, loft/lie modifications to customize clubs

SKILLS

Programming Languages

- C/ C++
- Verilog / System Verilog
- Python

Protocols & Interfaces

- I²C
- SPI
- UART

Software & Simulation

- LTSpice
- KiCad
- MatLab

Hardware Platforms

- Nexys A7 FPGA
- TI TM4C123
- 8051
- ESP32

IDEs

- Keil µVision
- VS Code
- Git Version Control
- Xilinx Vivado

Debugging

- Logic Analyzer
- Oscilloscope
- In-Circuit Debugger
- Serial Communication